

Rain or Shine? Optimal Utility Pricing under Different Weather Patterns

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Urban water utilities face conflicting objectives: recovering fixed costs, enforcing resource conservation, and maintaining equity under climate-driven weather volatility. Using household billing records matched with satellite-derived vegetation indices in Austin, Texas, I estimate residential demand under increasing block pricing via a discrete-continuous choice model. Counterfactual Ramsey-style pricing reveals that price alone cannot resolve these trade-offs during droughts: conservation constraints generate an average welfare loss of \$74 monthly for low-income households due to the prevalence of high-volume users across all income tiers. Complementary lawn-zeroscaping policies offset this burden, improving distributional equity where price mechanisms fail. Keywords: Ramsey Pricing, Multi-part Tariff, Demand Volatility, Distributional Effect

I. Introduction

The increasing variability of weather patterns, exacerbated by climate change, poses a significant challenge to the pricing of essential utility services, particularly water. Urban

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water utilities face a complex balancing act, primarily managed through their rate structures: they must ensure financial sustainability through cost recovery, promote resource conservation amid dwindling supplies, and maintain equitable access for all consumers, especially low-income households. This study focuses on these challenges within the context of Austin, Texas—a rapidly growing metropolitan area known for its highly variable precipitation. The primary objective of this paper is to develop and empirically implement an optimal water pricing framework, grounded in Ramsey principles and tailored for an Increasing Block Price (IBP) tariff, that explicitly accounts for and remains resilient while achieving policy goals related to financial feasibility, conservation, and distributional equity under extreme weather patterns.

The theoretical underpinning for many utility pricing schemes stems from Ramsey (1927), who proposed “second-best” solutions to cover large fixed costs while minimizing societal welfare loss, typically through an inverse elasticity rule. In practice, utilities often adopt IBP tariffs, where prices rise with consumption, to encourage conservation among high-volume users while ensuring affordability for basic needs. However, the efficacy of IBP structures is predicated on the assumption that high consumption correlates with high income and low price elasticity—an assumption severely tested by unpredictable weather. A critical operational constraint is that utility prices are typically preset for an entire year, lacking the flexibility to adjust to monthly or seasonal weather variations. This means a single, static IBP structure must bear the burden of achieving all policy goals, from mitigating excessive usage during dry months to offsetting revenue loss during wet months, all while addressing distributional concerns. This paper first offers reduced-form evidence of consumer sensitivity to precipitation and then structurally estimates demand under nonlinear IBP using a Discrete/Continuous Choice (DCC) model.

Previous literature has explored the complex nature of estimating public utility demand and the welfare impacts of IBP optimization. Hewitt and Hanemann (1995) and Olmstead, Hanemann and Stavins (2007) use DCC models to estimate price elasticity in urban water demand. Some studies, such as Castro-Rodríguez, María Da-Rocha and

Delicado (2002) and Nataraj and Hanemann (2011), have shown that demand is responsive and that specific IBP changes can generate welfare improvements. In particular, Szabo (2015), examining South Africa’s free water policy, demonstrated that nonlinear pricing alone can improve welfare and conservation goals without relying on ad-hoc subsidies. Though these studies have demonstrated that IBP can meet the conservation policy while maximizing consumer and producer surplus, the distributional effect of IBP has been less discussed, likely due to the lack of granular data. Recently, with more detailed data, research has been highlighting the potential shortcomings of relying heavily on IBP as a policy instrument, particularly regarding the distributional goal. Borenstein (2009) and Ito (2014) found that consumers appear to react more to average prices than to the marginal prices of multiple tiers, and households that usually have less understanding of the complex pricing structure are likely to have lower income levels. Ruijs (2009) and Echeverri (2023), while acknowledging that consumers respond to IBP changes, directly evaluate the distributional effect of IBP restructuring and concluded that higher-income households usually gain higher welfare through IBP price changes.

This paper fills a gap in the literature by exploring the effectiveness of IBP as a price instrument under the strain of different weather patterns. I align closely with Wolak (2016), who employed a Ramsey-style model to determine an optimal IBP structure for welfare maximization. However, to my knowledge, this paper is the first to develop and empirically apply a Ramsey-style optimal pricing model that uniquely incorporates different exogenous weather patterns—both in terms of mean shifts and variance shifts—to “stress test” the efficacy of IBP, and to prepare the utility’s policy goals under all types of weather conditions. Furthermore, the demand estimation is enhanced by the novel integration of high-resolution ($10\text{m} \times 10\text{m}$) satellite-derived Normalized Difference Vegetation Index (NDVI) data. My descriptive analysis and the structural DCC model reveal that, contrary to common assumptions, high-quantity users exist across all income strata and that higher-income households can exhibit greater price elasticity, particularly when weather pushes the demand curve rightward. This finding introduces significant challenges for the utility relying solely on IBP to achieve its distributional goals when

weather pushes the demand curve to the right, while conserving resources.

This research also relates to the industrial organization literature on firms' responses to weather-induced demand volatility, such as Lin, Schmid and Weisbach (2017) and Baumgartner et al. (2022), and to research about the social consequences of real-time electricity pricing by Holland and Mansur (2008) and the heterogeneities of social marginal cost caused by weather by Borenstein and Bushnell (2022). The latter two papers about electricity pricing also discussed the utility's conservation policy goal under volatile demand, but they focused on how frequently pricing should change in response to volatile demand, a unique feature that is more feasible in electricity pricing.¹ This paper takes the inflexibility of price change within a year as an exogenous political constraint, and extends the existing paper by adding the specific concern regarding the distributional effects of pricing, on top of the environmental concerns, by exploiting the weather-induced heterogeneity of urban residential water demand.

Lastly, this paper contributes to a growing literature evaluating the distributional effects of climate-related policies (Parry and Williams III (2010), Goulder et al. (2019), Känzig (2023)) in both the field of industrial organization and macroeconomics, which has often found such policies to be regressive. Instead of targeting a specific green policy, this paper extends this strand of literature by evaluating how much of the regressiveness could be controlled by the utility's policy goal, and how much of it is exogenous and caused by weather. This disentanglement allows me to quantify the shadow cost of the conservation policy goal towards lower-income households and the trade-off between the policies of conservation and the distributional effect through price under various weather patterns.

I find that when weather pushes the demand curve to the right (e.g., drought), price alone struggles to improve, or even maintain, the distributional effect due to the binding conservation constraint. When demand shifts rightward due to drought or high weather variance, the conservation constraint imposes an average welfare loss of \$74.2 and \$74.9

¹Due to existing industry norms (like peak and valley pricing), consumers are more likely to accept real-time electricity pricing, and the utility will face less political consequences, as oppose to water utility.

per household per month for the lowest-income stratum, respectively. However, when demand shifts leftward due to more precipitation or low weather variance, the optimal price helps the lower-income stratum gain an average welfare of \$45.3 and \$48.5, respectively.² The estimated shadow cost of the conservation policy during drought can inform the design of complementary policies to achieve distributional objectives. For example, a 30% household zeroscaping/xeriscaping³ effort could generate \$72.07 in welfare per household per month for the lowest-income stratum, almost nullifying the welfare loss from the conservation policy's shadow cost.

II. Water Utility Pricing

In this section, I introduce water utility pricing in Austin, TX, and detail the challenges water utilities generally face under volatile weather conditions. This includes the current pricing structure and the policy goals utilities set to achieve, noting in particular that the price structure is often predicated on an assumed correlation between household income and consumption levels/price elasticities. Subsequently, I present the datasets used and examine whether these assumptions could hold, and spotlight the challenges faced by the utility's pricing decision under volatile weather.

A. Utility Pricing and Increasing Block Pricing (IBP)

A utility company, typically a natural monopoly with large fixed costs, faces a classic challenge outlined in Ramsey's pricing problem (from Ramsey (1927)): pricing at marginal cost, though efficient, would not cover its total costs, rendering the "first-best" solution infeasible. Ramsey proposed a "second-best" solution to meet the company's financial requirements while minimizing the reduction in social welfare. This price-setting solution is known as the inverse elasticity rule, where prices are higher for consumers with lower price elasticity, as inelastic demand causes smaller reductions in consumption and thus smaller welfare distortions. In the case of a water utility, where major

²This is due to the less demanding revenue goal set by Austin Water. A different utility with a more pressing revenue-feasible requirement could also cause welfare losses for the lower-income households.

³An effort to remove water-intensive vegetation to save more water.

costs are fixed (e.g., reservoir management, purification, and distribution infrastructure) while the marginal cost per household is relatively small, Ramsey pricing helps ensure the utility's financial viability while maximizing consumer welfare.

In addition to ensuring cost recovery, the utility must also meet its policy goals. Typically, the utility must consider two major policy objectives: 1) **resource conservation** and 2) the **distributional effect** of the price. With limited natural resources, the utility is encouraged to reduce total consumption—a constraint that has become more pressing with dwindling supply caused by climate change. On the other hand, the utility must also ensure that water remains affordable for low-income consumers' basic needs. Therefore, the optimal price must strike a balance: it cannot be so high as to be unaffordable for low-income households, nor so low as to jeopardize revenue and encourage excessive consumption.

A common practical solution is to adopt a multi-part tariff with Increasing Block Pricing (IBP), which categorizes consumers into tiers based on their consumption. For all tiers, the final bill consists of a fixed charge (or access fee) to cover fixed costs and a volumetric charge (or usage fee) based on consumption to cover marginal costs. In an IBP structure, both the fixed and volumetric charges typically increase with higher tiers. For the volumetric charge, consumers pay a lower marginal price on consumption up to each kink point and a higher marginal price only on the amount exceeding it (See Table 1 below for the pricing structure in Austin). This structure encourages high-quantity users to conserve, while lower-quantity users can benefit from a low marginal price (often below marginal cost). Predicated on the assumption that higher-quantity users typically have higher incomes and lower elasticities, IBP can be interpreted as an empirical application of Ramsey pricing that also addresses the policy goals of conservation and equity.

For this paper, I am going to specifically examine the efficacy of IBP in the context of various weather patterns. For the electricity utilities, pricing with the existing policy goals would be difficult to balance under unexpected heat waves, in addition to regular seasonal demand cycles. For the water utilities, volatile weather will cause complexities in pricing decisions as well. Urban household water usage is highly correlated to pre-

Table 1—: Pricing Structure

Kink Points (kGallon)	Fixed Charge (\$)	Marginal Price (\$)	
		Non-CAP	CAP
0-2	8.5	3.09	2.42
2-6	10.8	5.01	4.1
6-11	16.5	8.54	6.72
11-20	37	12.9	11.56
>20	37	14.41	14.26

Note: Each fixed charge is composed of fixed payment and meter charge. The fixed payment depends on final consumption quantity, and the meter charge assumes a 5/8 meter size, which is the most common residential meter size. The marginal price includes volume charge, reserved fund charge and community benefit charge. These itemized charges are all billed per 1,000 gallons, meaning they are essentially part of the marginal price. Volume charge depends on the amount of final consumption quantity while the other two are billed per 1,000 gallons. Reserved fund surcharge goes into a restricted reserve fund to offset water service revenue shortfalls that may impact operations and services. Community benefit charge is only billed to Non-CAP consumers to fund the CAP.

cipitation for single-family homes, where lawn watering constitutes a major portion of water consumption, far larger than essential usage like cooking, showering, and washing. Low precipitation (dry months) encourages much higher water usage, while high precipitation (rainy months) reduces usage as households may not need to use extra water for lawn watering. Therefore, for cities like Austin, where high precipitation events can occur randomly throughout the year⁴, it is important to design the IBP with the policy goals of conservation and maintaining financial viability to prepare for all weather situations, and the results derived from this paper can be applied to public utilities in general, as both electricity and water utilities have similar policy goals and unpredictable demand through volatile weather.

The importance of setting the price for all weather conditions creates another price-setting challenge for utilities, as the price typically cannot change from month to month. While economists typically suggest dynamic pricing, such as that used in ride-sharing services, to combat volatile demand, in the case of water utility pricing, the price must be preset for the entire year and cannot change within a certain time period. This inability to adjust prices in response to weather stochasticity is specifically what this paper

⁴For Austin, and many cities in Texas, high rainfall months could happen in any months between March to June, and September to November. These months could equally have low rainfall due to some years with a longer summer season.

addresses.⁵

B. Water Utility Pricing in Austin, TX

For this paper, I use water utility transaction data from Austin, TX. The water utility in Austin is managed solely by Austin Water, a public entity and natural monopoly. All operations—from water supply management in Lake Travis to purification, transport, wastewater processing, and price setting—are managed by Austin Water. This means Austin Water acts as a social planner aiming to maximize welfare while pursuing policy goals, making a Ramsey pricing framework highly suitable for analysis.

Austin, and Texas in general, are known for unpredictable precipitation. A report by Nielsen-Gammon et al. (2020) noted that future precipitation trends in Texas are likely to be dominated by largely unpredictable natural variability and a projected increase in the intensity and frequency of extreme rainfall events. This makes the price-setting challenge more acute, as it is difficult to predict which months will be wet or dry, rendering preset price discrimination based on seasonality nearly impossible. In Austin, as shown using NOAA data in the 'Precipitation Trend in Austin' section of the Online Appendix, recent precipitation has become more erratic compared to the 30-year average. This trend, combined with Austin's rapid population growth, makes residential water conservation an increasingly pressing issue.⁶

The problem faced by Austin Water is not unique. Many U.S. cities face the combined pressures of volatile weather, growing populations, and dwindling natural water supplies due to climate change.⁷ This paper aims to test whether a Ramsey-style pricing solution can achieve the policy goals of conservation and distributional equity under volatile weather. I utilize a panel dataset from Austin Water for approximately 120,000 households from May 2018 to December 2019, public data from the Travis County Appraisal District, weather data from NOAA, and high-resolution (10m × 10m) satellite imagery

⁵Implementing dynamic pricing in water utilities lacks industry precedent. Additionally, unpredictable seasonal precipitation makes it politically and practically difficult to establish proactive, stable monthly rate structures like those used in electricity.

⁶According to recent census data, the Austin metro area is one of the fastest-growing regions in the U.S.

⁷The EPA confirms that hourly rainfall rates and the intensity of heavy precipitation events have increased across most of the U.S. since the 1970s.

capturing household vegetation health. These data sources are detailed in Section III.B.

Austin Water’s current pricing uses a five-tier IBP structure, detailed in Table 1. This design aims to improve distributional outcomes and encourage conservation by setting higher marginal prices and fixed payments for higher-quantity users. This is based on the assumption that higher-quantity users are correlated with higher income levels, have lower price elasticities, and can afford higher prices. To better measure distributional effects, I segment households into five income strata, aligning with the five-tier price structure, to analyze how policies affect different economic groups (see the ‘Tier and Income Distribution’ section of the Online Appendix).

C. Descriptive and Reduced-Form Evidence

To study the impact of volatile weather on the utility’s price-setting problem and the challenges its policy goals present, I will first test the assumption of a correlation between income and quantity.

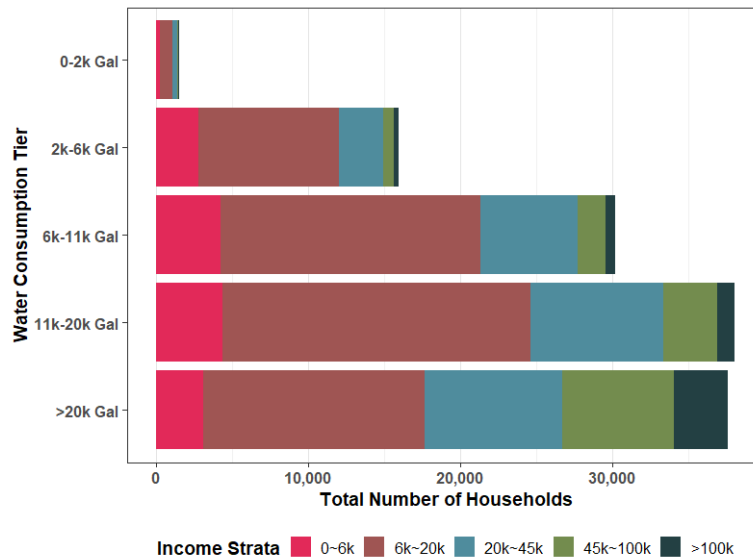


Figure 1. : Households Income Strata Composition for Each Tier

Figure 1 showcases the distributions of the average monthly quantity for each house-

hold, and the composition of the income strata for each tier. There are more households from the highest stratum that consume in the highest tier ($> 20k$ Gal). However, more than 3000 households from the lowest strata consume on average in the highest tier, which should be mostly occupied by households from higher strata. This undermines the premise that lower marginal prices in lower tiers effectively target low-income households for distributional benefits, and this omnipresence of high-quantity consumers across all income strata poses a challenge for using IBP to achieve equity goals.

In addition, to understand the impact of weather, I examine the correlation between weather changes and quantity changes from May 2018 to December 2019. To account for seasonality, I calculated the monthly difference in precipitation (Δ Precipitation) and temperature (Δ Temperature) relative to their 30-year averages (1990-2020). Using panel data, I then calculated the percentage deviation of a household's monthly consumption from its historical average for that same month. I am particularly interested in how monthly weather deviations influence this quantity deviation and how the effect differs across income strata.

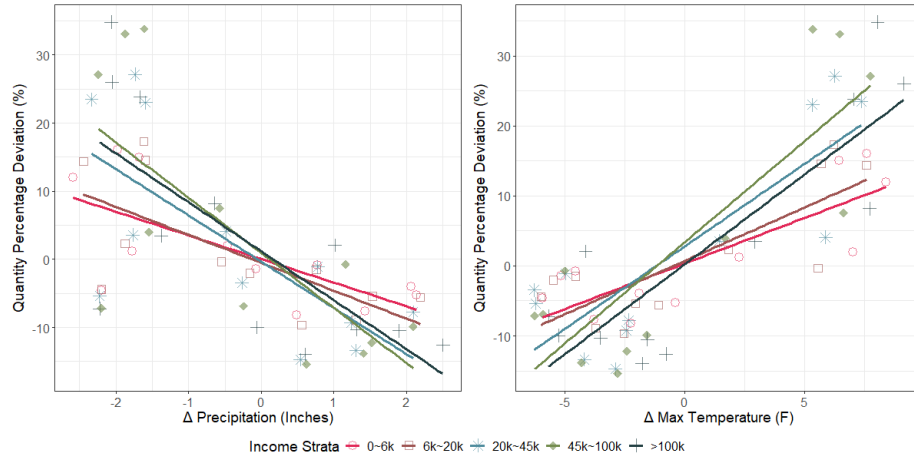


Figure 2. : Quantity Change (%) by Weather Deviation for each Income Strata

We can see that higher-than-usual precipitation typically leads to lower-than-usual consumption. However, the percentage deviation is greater for high-income households,

likely reflecting higher demand for outdoor water use. A similar, though less pronounced, effect appears to exist for temperature. A reduced-form analysis of Δ Precipitation, Δ Temperature, and their interactions with income strata on quantity deviation (see 'Descriptive and Reduced Form Evidence' section of the Online Appendix) confirms these patterns.

Based on the results, higher income strata tend to have a lower baseline percentage deviation from their usual consumption (when weather differences and marginal price are zero) compared to the lowest income group. However, higher income strata ($> 6k$) show a stronger negative response to increases in precipitation difference. The linear response to temperature difference is less clear-cut across strata in this model; while the base effect for the reference group is positive, higher income groups show interactions that temper this positive effect, resulting in less positive linear sensitivity compared to the low-income strata. Due to this unclear effect from the reduced-form analysis, coupled with the intrinsic correlation between precipitation and single-family home water usage, I will focus on the impact of changing precipitation for the rest of the paper and observe the shifts in consumer behavior.

This heightened weather sensitivity among higher-income households is somewhat counterintuitive and challenges the assumption that they have uniformly low price elasticity. This creates potential complexities for designing an optimal pricing structure that can meet all policy goals under volatile weather. To accurately evaluate counterfactual prices and revenue risks, it is crucial to measure how behavior changes in response to price, precipitation, income, and their interactions. Therefore, a structural model is needed to incorporate the full vector of household characteristics—a necessity given the limited price variation in the dataset—followed by an empirical Ramsey model that considers weather stochasticity to estimate the optimal IBP.

III. Demand Model and Estimation

In this section, I develop the demand model for the water utility to estimate the parameters needed for optimal price calculation and counterfactual analysis. As noted

earlier, residential water demand typically faces an IBP structure, meaning that the nonlinear budget constraint generated must be accounted for. Therefore, I utilize a Discrete/Continuous Choice (DCC) model estimated via Maximum Likelihood to estimate the conditional demand probabilities for each pricing tier for each household, thereby accurately estimating the demand parameters.

Each household chooses between water consumption and a numeraire good subject to its budget constraint. The pricing scheme creates a nonlinear budget constraint, and I focus exclusively on the IBP structure in this paper. I assume a log-log functional form relating demand to prices and income, which is a common assumption in the water demand estimation literature.

The first paper to introduce a demand model handling piece-wise linear budget constraints arising from price nonlinearity was Burtless and Hausman (1978) in the context of labor supply. Dubin and McFadden (1984), Hanemann (1984) and Hewitt and Hanemann (1995) laid the groundwork for applying the DCC model to residential utility demand (with Hewitt and Hanemann (1995) specifically addressing water demand). Our approach largely follows this established model structure, with minor modifications, to maintain interpretability.

A. Demand Model

The demand model assumes each household consumes both water (w) and a numeraire good (Y) (with price = 1), and the household's total monthly income is I . Suppose for an IBP, the marginal price for each tier k is p_k , the fixed payment is A_k , and the cutoff point between tier k and $k + 1$ is q_k . Due to the nature of IBP and suppose there are a total of K tiers, $p_{k+1} \geq p_k$, $A_{k+1} \geq A_k$, $q_{k+1} \geq q_k \forall k \in \{0, 1, \dots, K\}$. Together $\{p_k, q_k, A_k\}_{k=1}^K$ fully identify the nonlinear pricing structure. Throughout this section, the demand for every month should have a subscript of t , but for simplicity, the time-level subscript will be omitted.

Condition on the household choosing the optimal tier to be $k \leq K$ and the starting price

$p_0 = 0$, the household's budget constraint is:

$$\begin{aligned}
 I &= A_1 + p_1 w + Y \quad (\text{if } k = 1) \\
 &= A_k + \sum_{j=1}^{k-1} (p_j - p_{j+1}) q_j + p_k w + Y \quad (\text{if } k > 1) \\
 p_k w + Y &= I - A_k - \sum_{j=1}^{k-1} (p_j - p_{j+1}) q_j
 \end{aligned}
 \tag{1}$$

Therefore, given the IBP pricing structure, the only way to maintain the typical utility maximization problem with a budget is to add the additional term on the income such that the Virtual Income is equal to $I + d_k$, where the correction term d_k is defined as:

$$\begin{aligned}
 d_k &= -A_1 \quad (\text{if } k = 1) \\
 d_k &= -A_k - \sum_{j=1}^{k-1} (p_j - p_{j+1}) q_j \quad (\text{if } k > 1)
 \end{aligned}
 \tag{2}$$

The concept of Virtual Income is used to correctly model consumer choice under a tiered pricing system. When a household's consumption is in a higher tier, k , the initial "blocks" of consumption (tiers 1 through $k - 1$) are purchased at lower marginal prices. To simplify the analysis, we can treat the consumer's choice problem as if they face a single marginal price, p_k , for all units. For this simplified model to yield the same utility-maximizing choice, the consumer's income must be adjusted upward to account for the savings on the initial, lower-priced blocks.⁸ To visually illustrate, consider a simpler model with 2 tiers without loss of generality:

Since consuming at a higher tier will grant paying the lower quantity portion at a lower marginal price, the correction term d_k (which will be > 0 in the higher tier given that $p_k \leq p_{k+1} \forall k$ and A_k is not too large) has an income effect on the overall budget set. This correction term will allow us to solve for the utility maximization problem using the marginal price for each tier without causing any trouble.

⁸This additional income, often denoted as d_k , can be considered an income effect. It allows the household to be modeled as if it faces the marginal price p_k across all quantities, even though the initial blocks were purchased more cheaply.

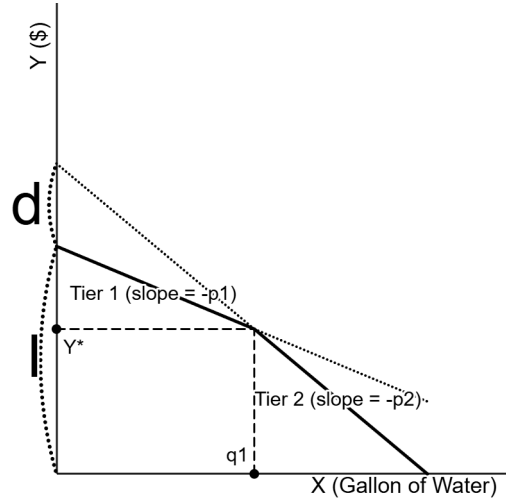


Figure 3. : Virtual Income $I + d_2$ when $K = 2$

Define the utility function for the household between water and numeraire good to be $U(w, I)$. Condition on tier k being the optimal choice for the household, define the conditional water demand function to be $g(p_k, I)$, where the functional form will be introduced later. Then the conditional indirect utility function for tier k is:

$$\begin{aligned}
 (3) \quad V(p_k, I) &= \max_w U(w, I - p_k w) \\
 &= U(g(p_k, I), I - p_k g(p_k, I))
 \end{aligned}$$

For general U and budget sets, there will be a case where the household will have multiple optimal tiers such that the conditional indirect utility function is not unique. However, fortunately, given the case of IBP, Hausman (1979)'s theorem showed that if the budget set is convex, the optimal tier will be unique. In addition, the concavity of the utility function U will make sure the optimal consumption point $g(p_k, I)$ resides in the specific quantity boundaries of each tier. Therefore, without loss of generality, the unconditional

indirect utility function for the total tier $K = 2$ is defined as:

$$(4) \quad V(p, I) = \begin{cases} V(p_1, I) & \text{if } q_1 \geq g(p_1, I) \\ V(p_2, I + d_2) & \text{if } q_1 < g(p_2, I + d_2) \\ U(q_1, I - A_2 - p_1 q_1) & \text{if } g(p_2, I + d_2) \leq q_1 < g(p_1, I) \end{cases}$$

The form of the unconditional indirect utility function can be expanded towards a more general case. I would like to highlight the case of $g(p_2, I + d_2) \leq q_1 < g(p_1, I)$ to make sure the Incentive Compatibility constraints are satisfied throughout the entire quantity line. The reason for the inclusion of this case is explained in detail in the 'Additional Case in the Unconditional Indirect Utility Function' section of the Online Appendix.

So far, I have covered the decision process of the household's water demand, in which they solve for a utility maximization problem and solve for $g(p_k, I)$ for all k , and choose the k that maximizes their utilities only with the assumption that the demand function is concave without imposing any functional form on the demand function. A typical functional form used by recent literature of water demand is a log-log demand function⁹, I will use the same approach as the data of water quantity w_i is very right-skewed with only positive values. Therefore, a better parametric assumption for the distribution of w_i should be log normal, which will induce a log-log demand function. With vectors of household characteristics X and weather Z , the conditional indirect utility function for tier k has the functional form of:

$$(5) \quad V(p_k, I + d_k, \beta) = -\exp(\beta'_1 X + \beta'_2 Z + c + \varepsilon) \frac{p_k^{1-\alpha}}{1-\alpha} + \frac{(I + d_k)^{1-\rho}}{1-\rho}$$

This functional form, combined with Roy's Identity, will give us the log-log demand form for tier k :

$$(6) \quad \log(g(p_k, I + d_k)) = \log(w_k) = \beta'_1 X + \beta'_2 Z - \alpha \log p_k + \rho \log(I + d_k) + c$$

⁹A couple of relatively recent examples of using this functional form for water demand are Hewitt and Hanemann (1995), Olmstead, Hanemann and Stavins (2007), and Wolak (2016). Others, like Szabo (2015) used a linear demand function.

where d_k is defined in Equation 2, α is the log price effect, ρ is the log virtual income effect, and c is the constant. The observed log consumption includes unobservables ε : $\log(w) = \log(w_k) + \varepsilon$. Following the same parametric assumption of Olmstead, Hanemann and Stavins (2007), the unobservables have two terms $\varepsilon = \eta + v$. $\eta \sim N(0, \sigma_\eta^2)$ represents the household-level heterogeneity observed by households (e.g., preferences), while $v \sim N(0, \sigma_v^2)$ represents the household-level perception/optimization error that is unobserved by households even ex-post. The idea is that it is nearly impossible for a household to precisely consume the water quantity (w_k) they would ideally demand given their preferences. For instance, there will always be extra cold water down the drain when she demands warm and hot water, for example. For all households, they are risk-neutral with $E[v] = 0 \forall k$. Neither the error terms are observed by the econometrician, hence the parametric assumption. In addition, this parametric assumption can allow us to derive a closed-form likelihood function for MLE for the observed water demand w_i for household i (omitting i subscript below for brevity):

$$f(w_i|X, Z) = \sum_{k=1}^K \left[\frac{1}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} \phi(s_k) (\Phi(r_k) - \Phi(n_k)) + \frac{1}{\sigma_v} \phi(u_k) (\Phi(m_k) - \Phi(t_k)) \right]$$

where:

$$(7) \quad \begin{aligned} t_k &= (\log(q_k) - \log(w_k)) / \sigma_\eta \\ r_k &= (t_k \sigma_\eta - \rho_s s_k \sqrt{\sigma_\eta^2 + \sigma_v^2}) / (\sigma_\eta \sqrt{1 - \rho_s^2}) \\ \rho_s &= \sigma_\eta / \sqrt{\sigma_\eta^2 + \sigma_v^2} \\ s_k &= (\log(w_i) - \log(w_k)) / \sqrt{\sigma_\eta^2 + \sigma_v^2} \\ n_k &= (m_{k-1} \sigma_\eta - \rho_s s_k \sqrt{\sigma_\eta^2 + \sigma_v^2}) / (\sigma_\eta \sqrt{1 - \rho_s^2}) \\ m_k &= (\log(q_k) - \log(w_{k+1})) / \sigma_\eta \\ u_k &= (\log(w_i) - \log(q_k)) / \sigma_v \end{aligned}$$

where Φ and ϕ are the standard normal cdf and pdf, respectively; and $q_0 = 0$, $q_K = \infty$, with $w_k = g(p_k, I + d_k)$ defined implicitly via Equation 6.

Empirically, this likelihood function is used to calculate the likelihood of the observed water consumption w_i given the model parameters $(\alpha, \rho, \beta'_1, \beta'_2, c, \sigma_\eta, \sigma_v)$ and data $(X_i, Z_i, p_k, q_k, A_k, I_i)$.¹⁰ The derivation of the likelihood function is explained in a 2-tier case in the 'Likelihood Function' section of the Online Appendix without loss of generality.

B. Empirical Model and Data

In this subsection, I outline the empirical model, based on the demand function (Equation 6) and the likelihood function (Equation 7), and describe the data used for maximum likelihood estimation.

The core dataset consists of panel data detailing monthly water transactions for single-family households in Austin, TX, from 2018-2019. These data include payments, total water usage, the billing date for each month, and the household's longitude and latitude. The data were provided by Austin Water, the public utility monopoly responsible for water services in Austin.

To operationalize the demand function, I incorporate supplemental datasets. These include two variables in the household characteristics matrix (X): the number of bathrooms per household and the time-variant household Normalized Difference Vegetation Index (NDVI). The number of bathrooms serves as a proxy for indoor water usage, while NDVI accounts for outdoor water usage, which is typically larger in volume and more sensitive to weather variations. The application of NDVI to water demand analysis was introduced by Wolak (2016). It aims to approximate lawn watering habits, as higher NDVI values indicate healthier vegetation, suggesting greater watering efforts by the household. Details on constructing and interpreting NDVI can be found in the 'NDVI' section of the Online Appendix. For the time-variant weather matrix (Z), I include the monthly average maximum daily temperature, the monthly interquartile range (IQR) of maximum daily temperature, total monthly precipitation, and the monthly IQR of total precipitation.¹¹

¹⁰In practice, I used $\alpha = \exp(\beta'_3 X_\alpha + \beta'_4 Z_\alpha + c_\alpha)$ and $\rho = \beta'_5 X_\rho + c_\rho$ to maximize the bias-variance trade-off and provide flexibility in identifying household heterogeneity in the price and virtual income effects.

¹¹Note that both IQRs account for the spread of the data within each month, which differs slightly from what I mean by growing weather variance, namely the increasing variance of weather patterns within a year.

To enhance model flexibility, I allow the log price effect (α) and the log virtual income effect (ρ) to depend on household characteristics and weather conditions, thereby generating interaction terms. Specifically, α depends on the number of bedrooms, household NDVI, monthly average maximum daily temperature, and total monthly precipitation. ρ depends on the number of bedrooms, NDVI, and the number of heavy water-use appliances (including pools, hot tubs, sprinkler systems, fountains, and car washes).¹²

All other household characteristics, except for NDVI, were collected from public data provided by the Travis County Appraisal District (TCAD). I use data from 2018 to match the time frame of the transaction data. The TCAD data also include addresses, which were matched via geo-spatial analysis with the longitude and latitude data from Austin Water. I also collected lot size data from TCAD. Weather data were collected from the National Oceanic and Atmospheric Administration (NOAA), using data from approximately 120 weather stations in Austin. Since NDVI and weather data are collected on a calendar month basis, while household water billing cycles do not typically align with calendar months, I prorate both the weather and NDVI data to match each household's billing cycle.¹³ Household income data were estimated using zipcode-level average homeowner income data from the IRS combined with normalized household value data from TCAD, under the assumption that the normalized variance of household values within each zipcode mirrors the variance of household income.

The final sample consists of 127,323 households with 2,345,742 transaction records. Summary statistics can be found in the 'Summary Statistics' section of the Online Appendix.

C. Estimation Result and Identification

After conducting the maximum likelihood estimation using Equation 7 with $k = 5$ and the data discussed in Section III.B, Given the maximum likelihood model using Equation 7 with $k = 5$ and the interaction terms, I estimate the model with a total of 18 parameters

¹²In practice, I used $\alpha = \exp(\beta_3'X_\alpha + \beta_4'Z_\alpha + c_\alpha)$, and $\rho = \beta_5'X_\rho + c_\rho$ to maximize the bias-variance trade-off of the estimation results and provide flexibility in identifying household heterogeneity in the price and virtual income effects.

¹³This practice was first introduced by Train et al. (1984).

$(\beta'_1, \beta'_2, \beta'_3, \beta'_4, \beta'_5, c, c_\alpha, c_p, \sigma_\eta, \sigma_v)$ from data $(X_i, Z_i, X_{\alpha,i}, Z_{\alpha,i}, X_{p,i}, p_k, q_k, A_k, I_i, w_i)$. All data measuring price or payment are scaled from nominal value to real dollar value in January 2017 using the Federal Reserve Economic Database Gross Domestic Product (GDP) deflator from St Louis Federal Reserve Bank.¹⁴ The estimation results are listed in Table 2:

Table 2—: MLE Estimation Results

Parameter Name	Estimate	Standard Error	Parameter Name	Estimate	Standard Error
Bathroom	1.16	(9.20E-04)	Average High Temp	0.00218	(2.04E-05)
NDVI	1.19	(0.011)	IQR High Temp	-0.0189	(1.46E-04)
Constant	0.731	(0.0017)	Total Precipitation	-0.941	(1.34E-04)
Price * bedroom	0.0486	(3.94E-04)	IQR Precipitation	0.12	(0.005)
Price * NDVI	-0.0404	(0.00465)	Income * Heavy Water Appliances	-0.0718	(4.81E-04)
Price * Avg High Temp	-0.0168	(2.47E-05)	Income * Bedroom	-0.0158	(5.34E-05)
Price * Total Prcp	-0.0363	(4.23E-04)	Income * NDVI	-0.0692	(3.91E-04)
Price	0.688	(0.00193)	Income	0.162	(1.67E-04)
σ_η	2.56	(8.45E-04)	σ_v	4.71E-04	(2.25E-03)

Note: Price * bedroom represents the interaction term inside α , measuring how the price effect is changed through the number of bedrooms. The same goes for other interaction terms for price and income.

Most results of MLE fit the intuition qualitatively. The standard error is calculated by the inverse of the matrix of the sum of the outer products of the observation-by-observation gradient of the log-likelihood for each household evaluated at the maximum likelihood parameter estimates, a method introduced by Hall, Hall and Hausman (1974).

With the panel data, both time-variant and time-invariant variables provide significant heterogeneity among households and across different months. This helps identify the parameters associated with household characteristics, income, and their interaction terms. Moreover, for weather variables, instead of using weather data for the entire city, I found the closest weather station for each household to introduce small variations, providing heterogeneity to help identify the weather parameters. Nevertheless, identifying the price effect and its interaction terms presents a challenge, as there is little or no price variation (neither through time nor geography) in the data. Another concern highlighted in past literature (e.g., Borenstein (2009); Ito (2014)) is that households may not react to the

¹⁴See Fred-GDPDEF

observed marginal price, as prices and the total payment amount are typically observed only after the billing cycle ends; instead, they might respond to the average price they face.

Fortunately, both of these identification concerns regarding price parameters in the DCC model have been addressed by Olmstead (2009). She demonstrated that each household, for each month, is optimizing over the entire price schedule. Consequently, the econometrician can recover the parameter estimates, along with the probabilities that households consume on each of their budget segments and at each kink point, directly from the DCC model results. In addition, the DCC model is estimated without ever determining the ‘observed’ marginal price of consumption – all the prices and the kink points enter the likelihood function, regardless of where consumption is actually observed. Olmstead (2009) also pointed out that price elasticity estimation has smaller bias when using the DCC model if variation in demand is driven primarily by household preferences (η) instead of the perception error (v), and our estimation results align with this requirement, showing that σ_η is much larger than σ_v . Furthermore, the model generates less bias if the existing price jumps between different tiers are sufficiently salient. Although there is no definite conclusion on how salient the price jump is needed to minimize the estimation bias, the marginal price differences between tiers in Austin are among the largest in the US¹⁵.

To further address the price elasticity identification challenge, I also utilize data from a small category of consumers enrolled in the Consumer Assistance Program (CAP). Based on certain income requirements, these consumers enjoy lower marginal prices (see Section 1). Although the variation is small, these consumers provide a small degree of price variation to aid in the identification of price elasticity.

Another potential endogeneity issue involves the use of NDVI, as the NDVI of a given calendar month could be the result of water usage in that same period. Therefore, I used a lagged term for NDVI, meaning that water demand this month is dependent on the

¹⁵See <https://bseacd.org/conservation-based-rate-structures/> for “Conservation-Based Rate Structures” - Barton Springs/Edwards Aquifer Conservation District

previous month's NDVI. The idea is that households decide how much water to consume this month based on the health of their lawn vegetation last month. A higher previous month's NDVI may indicate the household takes more care of its lawn, helping to account for variations in outdoor water usage.

Based on the identification strategy discussed above, I now discuss the price and income elasticity derived from the DCC model. Due to the nonlinear pricing structure, I cannot directly use α and ρ to calculate price and income elasticities. Instead, by following Olmstead, Hanemann and Stavins (2007), I use a simulation-based approach. For a small number ξ , I calculate how expected demand changes in response to a small change in the status quo marginal price (p_0), holding income (I) and other factors constant.¹⁶ The formula for price elasticity is as follows, where the same structure applies to income elasticity:

$$(8) \quad \text{Price Elasticity: } (E[g((1 + \xi)p_0, I)] - E[g(p_0, I)]) / (\xi E[g(p_0, I)])$$

The median price elasticity is estimated to be -0.395 . Compared to previous literature that uses the structural DCC model for Increasing Block Pricing (IBP) in developed countries, our estimate is within the range of Pint (1999)'s estimations ($-1.24 \sim -0.04$) and more strictly, Olmstead, Hanemann and Stavins (2007)'s estimations ($-0.59 \sim -0.33$). Our estimate is more inelastic compared to Hewitt and Hanemann (1995)'s ($-1.63 \sim -1.57$). Our estimate is also close to that of Olmstead (2009) (reporting -0.609 for a specific DCC model specification, using data from 11 cities with diverse weather patterns) and Strand and Walker (2005) ($-0.3 \sim -0.1$, using data from 17 cities in Central America, where water demand was primarily for indoor usage). The median income elasticity is estimated to be 0.112 , which is close to estimations from previous literature for developed countries, such as Olmstead, Hanemann and Stavins (2007) ($0.1786 \sim 0.1865$) and Olmstead (2009) (0.1865).

¹⁶ Additionally, due to the flexibility of the price elasticity function, I limit the sample for empirical analysis to the year 2019, as opposed to from demand estimation, I used data from Jun. 2018 to Dec. 2019. This approach helps avoid overestimating the more rainy summer months of 2018.

To further explore the heterogeneities of price elasticities across income strata, I plot the empirical density of the price elasticities for all 5 income strata.

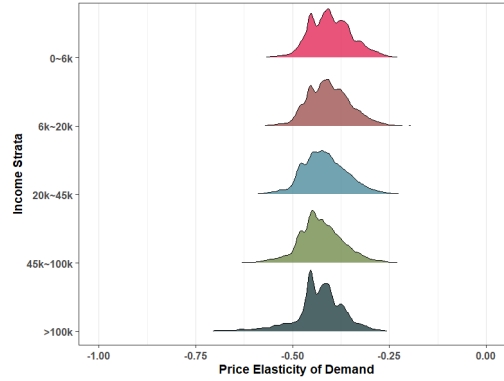


Figure 4. : Price Elasticity Density by Income Strata

There are very few heterogeneities of price elasticity across income strata, showcasing that the assumption that higher income households with higher consumption will have low elasticities is not correct, breaking the premise of using IBP to achieve the distributional goal. On the contrary, the highest income stratum even has a larger left tail, meaning the lower end of the highest income stratum is even more elastic compared to other strata. To further explain the reduced-form evidence from Figure 2, I separate the data into wet and dry months based on the median precipitation value.

There is a clear trend in the dry months where the median price elasticity is lower for higher income strata (larger in magnitude)¹⁷, meaning households with higher income are more responsive to price changes under low precipitation, while lower income strata do not share this difference. This counterintuitive result is likely due to the interaction between price effect and precipitation: when it rains a lot, few households across income levels have excess outdoor water consumption that responds to price, but when it is dry, households with the preferences of greener lawns and larger pool usage are more responsive to price change, and these households tend to have higher income. In addi-

¹⁷This fits the reduced-form evidence from Figure 2.

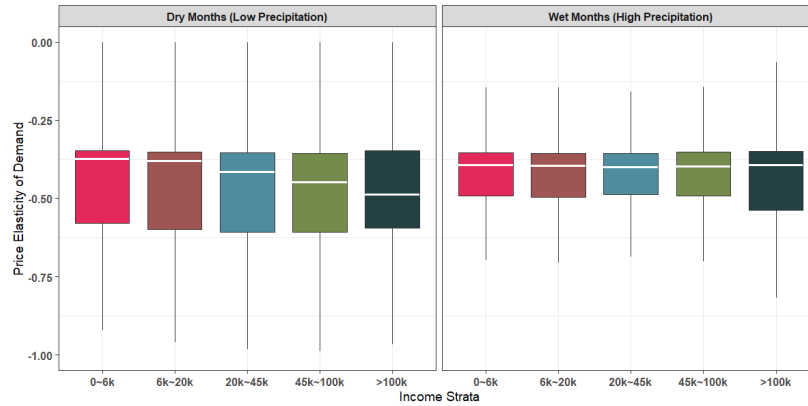


Figure 5. : Price Elasticity Boxplots by Income Strata - Wet vs. Dry Months

tion, higher-income households are more likely to be equipped with better technology to precisely control their water usage¹⁸, which will also contribute to higher price elasticities. From the figure in the 'Price Elasticities of Income Strata' of the Online Appendix, we can see that 14.4% of the highest income stratum households are more elastic (lower than the overall median), and consume on average more than 20k gallons.

This observation of high elasticities for higher-income households under dry weather, combined with the omnipresence of high quantity consumers across all income strata (see Figure 1), adds extra challenges to the optimal price design, especially in dry months. Under the inverse elasticity rule, the price should decrease for these households. However, a price decrease would result in an overblown increase in the consumption of the high-income, more elastic, and high quantity consumers, which poses a serious issue in reaching the conservation goal of the utility. In addition, a price decrease for high-income households will certainly create complexities in reaching the distributional goal of the optimal price. More discussions regarding the price elasticity heterogeneities among income strata can be found in the 'Price Elasticities of Income Strata' of the Online Appendix.

¹⁸Like smart meter, smart sprinkler, etc.

D. How weather shifts the Demand Curve?

From the structural demand model, I have estimated the heterogeneity of price elasticities across income strata and weather conditions, and observed that under dry weather, the higher income households would become more elastic, which creates complexities under Ramsey's inverse elasticity rule. This also showcases the importance of precipitation in the discussion of water utility pricing. In this subsection, I will introduce four alternative weather conditions and observe the changes in quantity and payments under the status quo pricing scheme to understand how weather shifts the demand curve. The four weather conditions are: 1) dry weather, where the precipitation decreases by 0.25 inches for every month compared to the status quo weather, 2) rainy weather, where the precipitation increases by 0.25 inches for every month compared to the status quo weather, 3) low variance, where the new precipitation standard deviation is $0.75\times$ the status quo standard deviation, and 4) high variance, where the new precipitation standard deviation is $1.25\times$ the status quo standard deviation. I only changed precipitation and kept all other weather variables unchanged. In addition, all other house characteristics remain the same as their status quo counterparts. I will then compare the new quantities and payments through the alternative weather and compare them to the status quo. I will present the overall distribution of the data through empirical cdf and then focus the data to the lowest income stratum ($0 \sim 6k$) and the highest income stratum ($> 100k$).

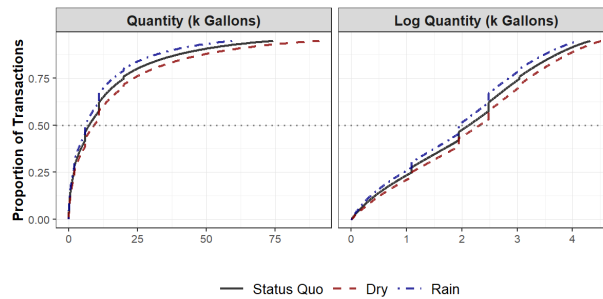


Figure 6. : Changes in Quantity and Log Quantity Distributions between Weather Conditions - Dry and Rainy

When the weather becomes drier, less precipitation will shift the demand curve rightward, causing quantity and payments to increase. Conversely, when the weather becomes more rainy, more precipitation will shift the demand curve leftward, causing quantity and payments to decrease.¹⁹ This means that in order to achieve a better conservation goal for the utilities, the status quo price is too low, at least in the higher tier, to better curb the excessive demand under dry weather conditions, while the financial feasibility concern is minimized. On the other hand, under rainy weather, the conservation goal is easy to achieve, while financial feasibility concerns are more pressing.

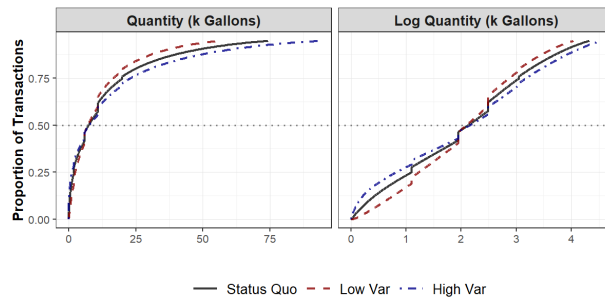


Figure 7. : Changes in Quantity and Log Quantity Distributions between Weather Conditions - Low and High Variance

When the weather decreases in variance, the demand distribution will have a smaller spread as well, meaning consumption left of the median will increase, and consumption right of the median will decrease. However, since both the precipitation and quantity are very right-skewed, the decrease in the consumption is larger than the mean to making it more dominant, which is evident when comparing the distribution between quantity and log quantity. Therefore, the empirical cdf largely resembles the pattern of more rainy weather, where the financial feasibility concerns are more pressing. Conversely, when the weather increases in variance, the demand distribution will have a larger spread, causing the consumption right of the median to increase. These shifts on the demand curve are more dominant due to the same reason, making the empirical cdf largely resemble

¹⁹The “bumps” in either empirical cdfs, are due to the consumer bunching around the kink points.

the pattern of drier weather, where the conservation goal is harder to meet. Comparing demand distributions across strata confirms that high-income households exhibit significantly larger “weather elasticity” under both shifting means and variances (see the ‘How Weather Shifts the Demand Curve’ section of the Online Appendix). Conversely, the relative inelasticity of low-income households poses a severe distributional challenge when weather pushes demand rightward. Under such conditions, binding conservation targets force price increases that inevitably generate shadow costs and reduce welfare. Given these counter-intuitive elasticities and the prevalence of high-volume users in lower-income brackets, it is doubtful that price alone can simultaneously achieve conservation and equity. Section IV formally defines the empirical optimal price model to test this limitation.

IV. Ramsey Pricing Model

In this section, I establish the theoretical basis for the Ramsey pricing problem (from Ramsey (1927)) faced by a natural monopoly water utility company with the policy constraints it is facing. Finally, I introduce the empirical model used to estimate the optimal Ramsey pricing under different extremes of weather conditions.

As previously established, Ramsey proposed the “second-best” solution to address a company’s financial requirements while minimizing the reduction in overall social welfare, employing the inverse elasticity rule. Water utility companies, particularly in urban metropolitan areas with high fixed costs, often adopt the Ramsey format of IBP within a multi-part tariff structure to meet financial requirements and policy goals of conservation and equity. For simplicity, when discussing the theoretical basis for the Ramsey problem, I treat the entire pricing structure as a single price p , as opposed to the pricing vector of IBP in a multi-part tariff. In the empirical model, I reintroduce the multi-part IBP to derive more realistic solutions.

Building on the generalized Ramsey framework Ramsey (1927); Coady and Drèze (2002), this empirical model adopts the conservation constraint structure from Wolak (2016) and measures welfare using equivalent variation under nonlinear budget con-

straints Hausman (1981); Reiss and White (2005); Ruijs (2009). While the formal Lagrangian—specifically how a binding conservation constraint mathematically alters the classic inverse-elasticity rule—is detailed in the ‘Ramsey Pricing Model’ section of the Online Appendix, this section focuses strictly on the empirical implementation.

A. Empirical Model

In this subsection, I develop the empirical model used to estimate the optimal price derived from Equation ???. The empirical model’s goal is to extend the Ramsey model into a more realistic setting.²⁰ To keep consistency with the demand model, the optimal price ($\mathbf{p}$) consists of three vectors: marginal prices ($\{p_k\}_{k=1}^5$), kink points ($\{q_k\}_{k=1}^4$), and fixed payments ($\{A_k\}_{k=1}^5$). To maintain Austin Water’s current pricing structure of 5 tiers with 4 kink points, $\mathbf{p}$ has a total of 14 parameters, and I assume the pricing structure does not change²¹. The optimal price, which depends on weather Z , maximizes the annual total equivalence variation (EV) weighted by household income I to boost the distributional effect.²² To account for weather-level stochasticity, ideally, I would need to utilize the real distributions of weather to derive E_Z . However, it is challenging to estimate this real distribution of weather consistently by using either data from very recent years (which lack sufficient data points for empirical distribution derivation) or long-term data (which are affected by climate change trends). Instead, I assume the utility faces an expectation based on its prior²³, and then calculates the optimal price based on its weather expectation. Later, I will perform a robustness check by adding a Monte-Carlo disturbance to simulate weather prediction error and generate a distribution of Z . Further details on simulating the distribution of Z can be found in the ‘Simulate Weather Distribution’ section of the Online Appendix. Aside from stochasticity from Z , the demand estimation using MLE (Equation 7) also introduces household-month-level

²⁰The model is partially based on Wolak (2016).

²¹This includes the assumption that each marginal price and fixed payment will be non-decreasing in tiers.

²²Following Feldstein (1972) and Atkinson (1970), welfare weight is set according to the social marginal utility of income. Furthermore, a logarithmic social utility of inequality and $\epsilon = 1$ for the Atkinson Index result in a weighting proportional to $\frac{1}{j}$.

²³In reality, utilities would form a general trend of next year’s weather based on phases of the El Niño-Southern Oscillation (ENSO). For example, next year’s weather will be drier in general, causing precipitation to be lower in the mean.

demand stochasticity from ε_h , as the econometrician observes neither η_h nor v_h . I use Monte-Carlo simulations for both to generate their distributions.²⁴

$$(9) \quad \begin{aligned} \mathbf{p}^*(Z) = \arg \max_{(\mathbf{p}, q, A)} \sum_h \left[\omega_h \cdot EV_h(Z; \mathbf{p}, \mathbf{p}_0, I) \right] - \lambda \cdot \max(0, C - R_h(Z; \mathbf{p})) \Big] \\ \text{s.t.} \quad P\left(\sum_h q_h(Z_h; \mathbf{p}) \leq \bar{Q}\right) \geq 0.95 \end{aligned}$$

Note that for the two exogenous thresholds: $C = \sum R(Z_0; \mathbf{p}_0)$ and $\bar{Q} = \sum q(Z_0; \mathbf{p}_0)$. Both Z_0 and $\mathbf{p}_0$ represent the status quo weather and price. This means I compare the counterfactual revenue and quantity to their status quo counterparts. In particular, due to the lack of detailed information on costs, I use the status quo revenue as the benchmark to evaluate financial viability. Utilities usually ensure their revenue is just enough to cover the annual cost to avoid excessive welfare distortion, which makes this a valid assumption. I further discuss the validity of this assumption in this case in the 'Status Quo Revenue as the Threshold' section of the Online Appendix. For the conservation constraint, I introduce chance-constrained programming to provide more flexibility in price setting. It essentially requires that the probability of the counterfactual annual total quantity for all households being smaller than the status quo annual total quantity is greater than 0.95. The structures of both constraints are adopted from Wolak (2016). Unlike the typical Ramsey pricing setting, I set up the revenue constraint as a revenue loss term $-\lambda \cdot \max(0, \text{loss})$, such that when the revenue requirement is not met, it incurs a cost to the entire economy, but does not necessarily award any positive value when the requirement is met. The goal is to capture the utilities' surcharge mechanisms or debt-service coverage rules that make large revenue losses more costly. The parameter λ controls the weight between the monetary value of welfare, measured by the weighted equivalent variation, and the loss of revenue for the utilities. Mathematically, it is similar to the Ramsey model, but empirically, this provides more flexibility as, instead of a hard constraint, the utilities are allowed to have some annual losses.

²⁴I will generate η_h on the household level to avoid overfitting.

The weight in front of the welfare, $\omega_h = I_{\text{median}}/I_h$ is to boost the distributional effect from the pricing optimization procedure. I took the inverse of the household income and centered the median to be 1, such that the welfare part and the revenue part are comparable. The weight is an assumption on the policy of the utility, as they will actively try to boost the distributional effect from the pricing. By making it such that higher than median income will receive smaller weights ($\omega_h < 1$) and lower than median income will receive larger weights ($\omega_h > 1$), the aggregated welfare will, through the optimization process, actively look for a price that can boost distributional effects.²⁵

The weight of the revenue loss λ is solved by searching on a grid, and estimating the optimal price under status quo weather.²⁶ I find that when $\lambda = 0.5$, it generates the result closest to the status quo prices. No further fine-grained grid search has been implemented, as I largely need to know the policy preference of the utility. $\lambda = 0.5$ means they weight the consumer welfare twice as important compared to revenue losses, which is reasonable for a social planner.

The welfare itself is defined as the equivalence variation ($EV_h(Z; \mathbf{p}, \mathbf{p}_0, I)$). It is calculated by obtaining the expenditure function from the indirect utility function and then calculating the welfare effects from the price change. The framework is developed by Hausman (1981) and Reiss and White (2005), and the formal definition is from Ruijs (2009).

Based on the indirect utility function $V(\mathbf{p}, I + d_k)$ from Equation 3, the expenditure function $e(\mathbf{p}, u)$ for a household choosing tier k is:

$$(10) \quad e(p_k, u) = \left[(1 - \rho) \left(u + \exp(\beta'_1 X + \beta'_2 Z + c + \varepsilon) \frac{p_k^{1-\alpha}}{1-\alpha} \right) \right]^{\frac{1}{1-\rho}}$$

If a household's counterfactual demand falls within tier k , rather than at a kink point

²⁵The idea of using income inverse as the base for social welfare calculation was first developed by Atkinson (1970). When $\varepsilon = 1$, the Atkinson index is $1/I$. The normalization of the weights such that a household with the median income receives a weight of 1 is a standard approach in applied cost-benefit analysis (e.g., Office of Management and Budget, 2023).

²⁶The grid searched upon is $\lambda = [0.25, 0.5, 0.75, 1, 1.5, 2, 5, 10]$.

q_k , its equivalent variation is:

$$(11) \quad EV(p_0, p, I) = e(p, V'(p)) - d_k^0 - I$$

where $V'(p)$ is the new counterfactual utility generated by the new price p , and d_k^0 is the virtual income correction term from the original price scheme, i.e., $d_k = -A_k - \sum_{j=1}^{k-1} (p_j - p_{j+1})q_j$. The equivalent variations are adjusted for the amount of subsidies received due to the nonlinearities in the budget set. Note that the expenditure function ($e(p_k, u)$) is used to solve for the virtual income from the indirect utility function, instead of I , hence there is no correction term d_k here. If the price does not change, $EV(p_0, p_0, I) = 0$ and $e(V_0, p_0) = I + d_k^0$.

On the other hand, when the predicted consumption is at a kink point q_k , the above definition does not apply as these consumers are not technically facing the new marginal price from their chosen tier.²⁷ The equivalent variation is generated from $\bar{p}$, where $\bar{p}$ is the price at which the demand function (from Equation 6) generates the result q_k . If I denote $\mathcal{A} = \beta_1 X + \beta_2 Z + c + \varepsilon$, and $\bar{V}I$ denotes the corresponding virtual income, the idea is to solve for $(\bar{p}, \bar{V}I)$ as a substitution effect where $(\bar{p}, \bar{V}I)$ generates the same utility.

$$\text{This means: } \begin{cases} V(p_k, I + d_k^0) &= V(\bar{p}, \bar{V}I) \\ \log(q_k) &= \mathcal{A} - \alpha \log(\bar{p}) + \rho \log(\bar{V}I) \end{cases} \quad . \text{ I can solve for } \bar{p}:$$

$$\log(q_k) = \mathcal{A} - \alpha \log(\bar{p}) + \frac{\rho}{1-\rho} \log[\exp(\mathcal{A}) \frac{1-\rho}{1-\alpha} (\bar{p}^{1-\alpha} - p_k^{1-\alpha}) + (I + d_k^0)^{1-\rho}]$$

For this demand function and indirect utility function, I need to solve this nonlinear function numerically to obtain $\bar{p}$. Then, the equivalent variation for predicted demand at the kink point q_k is:

$$(12) \quad EV(p_0, p, I) = e(\bar{p}, V'(p)) - (\bar{p} - p_0)q_k^0 - d_k^0 - I$$

²⁷To see more details of the additional case, see the 'Additional Case in the Unconditional Indirect Utility Function' section of the Online Appendix.

V. Counterfactual Analysis

In this section, I analyze the welfare effects of counterfactual optimal prices under various precipitation patterns, focusing on the differences in these effects across income strata. I quantify the associated welfare by comparing the new and status quo values for revenue ($R\%$), quantity ($Q\%$), and the equivalent variation to income ratio (EV/I). These measures capture the changes from both the price structure (including its policy constraints) and the weather itself. The results indicate that the lowest-income stratum is the most susceptible to welfare decreases from both exogenous weather shocks and the policy constraints within the price optimization process. I then disentangle the welfare change from these two sources to calculate the shadow cost of the policy constraints on the lowest-income stratum. Lastly, I propose zeroscaping/xeriscaping as a policy solution to specifically improve the distributional outcome.

To generate the counterfactual weather scenarios, I begin with the status quo precipitation (Z^0) and create two types of patterns: 1) Mean Shift: $Z' = Z^0 \pm \zeta_1$, where $\zeta_1 \in [-0.25, 0.25]$. The generated Z' has the same variance as Z^0 but a different mean. 2) Variance Shift: $Sd(Z') = Sd(Z^0) \cdot \zeta_2$, where $\zeta_2 \in [0.75, 1.25]$. The generated Z' has the same median as Z^0 but a different variance. The latter is achieved by nonparametrically scaling the standard deviation from Z^0 while maintaining the median and clipping the result at 0.²⁸ I focus solely on changes in precipitation, assuming all other weather variables remain at their status quo levels. Throughout the analysis, I also hold household characteristics and income constant, with the exception of NDVI, which is updated through a reduced-form analysis as discussed in the 'Counterfactual NDVI' section of the Online Appendix.²⁹ Using these counterfactuals, I employ the empirical model from Equation 9 to calculate the optimal price. Further details on the optimization procedure can be found in the 'Optimal Price Procedure' section of the Online Appendix.

²⁸ Although precipitation often follows a right-skewed log-normal distribution, a parametric assumption makes scaling by the factor ζ_2 imprecise. As noted by meteorologists Heredia et al. (2018), a nonparametric approach is more realistic for generating synthetic high-variance data.

²⁹ I also assume the income effect in the demand function (Equation 6) does not change with weather.

A. Welfare Results - Mean Shift (ζ_1)

This subsection presents the results of shifting the mean of precipitation by adding $\zeta_1 \in [-0.25, 0.25]$ to the status quo weather, showcasing the heterogeneous welfare effects across income strata, particularly when demand shifts rightward. Under this scenario, the utility expects the average precipitation to change by ζ_1 while the variance remains constant. The resulting menu of optimal prices can be found in the 'Price Result - Mean Shift' section of the Online Appendix.

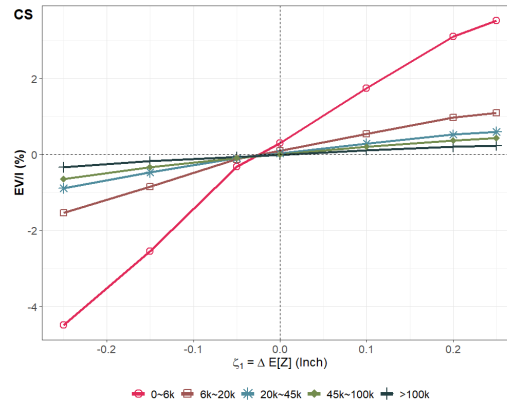


Figure 8. : Welfare from the Optimal Price in Shifting Mean ($\zeta_1 \in [-0.25, 0.25]$)

Figure 8 shows the general welfare results for precipitation expected to be drier ($\zeta_1 \leq 0$) and rainier ($\zeta_1 \geq 0$). When $\zeta_1 \leq 0$, demand shifts rightward, the quantity constraint becomes binding, and the revenue constraint becomes slack. Even with price increases only for higher tiers (see the figure in the 'Price Result - Mean Shift' section of the Online Appendix) and weighting the objective by EV/I , lower-income households exhibit a larger welfare loss compared to strata above \$20k, which show minimal loss. Conversely, when $\zeta_1 \geq 0$ (i.e., when the weather is rainier), demand shifts leftward, the revenue loss is more pressing, but due to the implementation of the loss function and lower weight ($\lambda = 0.5$), I still observe revenue losses. This results in a welfare increase for lower-income households.

One of the caveats from this analysis is that I don't directly observe the household-level

income. Instead, the income is extrapolated through the house value and normalized by zip code, assuming the distributions of house value and income within each zip code are the same. To test the validity of this assumption, I limited the welfare result to 1) the top 50th percentile of house built year (built after 1982) and 2) houses built after 2000 ($\sim 27\%$ of the total households), and the result could be found in the 'Welfare Results of Different Built Years - Robustness Check' section of the Online Appendix. To summarize, in both specifications, the lowest stratum still generates much lower welfare compared to other strata, and both results are at a similar level to the result from the full sample.

B. Welfare Results - Variance Shift (ζ_2)

This subsection presents the results of shifting precipitation variance (ζ_2), demonstrating how changes in variance also shift the demand curve. Based on the result from Section III.D, lower variance ($\zeta_2 < 1$) will shift the demand curve to the left, while higher variance ($\zeta_2 > 1$) will shift the demand curve to the right. Under this scenario, the utility expects the variance of precipitation to change by a factor of ζ_2 while the median remains constant.

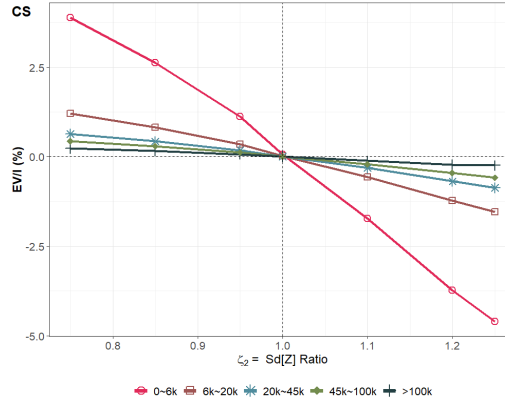


Figure 9. : Welfare from the Optimal Price in Shifting Variance ($\zeta_2 \in [0.75, 1.25]$)

Figure 9 showcases the general welfare results for $\zeta_2 \leq 1$ and $\zeta_2 \geq 1$. When $\zeta_2 \leq 1$

(variance decreases), demand shifts leftward, the revenue loss is more concerning, and the conservation constraint is slack, but again, due to the implementation of the loss function and $\lambda = 0.5$, I still observe revenue to be lower than the status quo when precipitation variance is very low. This occurs because, as variance decreases, the right-skewed precipitation distribution results in more rain during historically drier months. Consequently, welfare generally increases, particularly for the lowest-income stratum ($0 \sim \$6k$). Conversely, when $\zeta_2 > 1$ (variance increases), demand shifts rightward, the quantity constraint binds, and the revenue constraint is slack. The resulting different weather patterns decrease welfare for all strata, disproportionately affecting lower-income households (similar to the $\zeta_1 \leq 0$ case). The interplay of equity welfare weight (ω_h), both constraints and weather, creates an interesting pattern for quantity. When precipitation variance is low, to reduce revenue loss, the utility will “prefer” more quantities (ceiling by the upper bound) to be consumed by higher income households. However, when precipitation variance is high, the pressure to maintain the conservation goal is largely achieved by the decreasing quantity for the higher-income households. As the equity weight is larger for the low-income households, they don’t have the same “burden” as the high-income households to maintain the distributional goal.

Combining these results, a clear pattern emerges: the distributional effect of IBP is not improved, and is often worsened when weather shifts demand rightward, even with income-weighting in the objective function. Several factors explain this outcome: 1) Water bills naturally represent a higher proportion of income for lower-income households, so any price or weather change incurs a larger relative welfare shift (EV/I).³⁰ 2) The inverse elasticity rule suggests decreasing prices for high-elasticity consumers, which is distributionally counterproductive, as many high-elasticity households are in the highest income stratum and consume large quantities.³¹ 3) The most important reason is that a non-negligible number of lower-income households are high-quantity consumers

³⁰Similar patterns have been pointed out by Ruijs, Zimmermann and van den Berg (2008) using data in Sao Paolo, Brazil.

³¹See Figure 4 and Figure Price Elasticities v. Mean Quantity by Income Strata in the ‘Price Elasticities of Income Strata’ section of the Online Appendix.

with relatively low elasticities, often placing them in higher price tiers than their income suggests.³² As I explained above, the optimization process has tried its best to shift the burden of the conservation goal towards higher-income households, and we see a large decrease in quantity when demand shifts to the right. However, the process is only achieved through targeting quantity, instead of targeting income. The existence of high-quantity-low-income households will inevitably suffer from higher prices from the conservation constraint, which leads to lower welfare. Even though they are not the majority of the lowest income stratum, they still drag the results down.

All these reasons compromise the premise that income is a good indicator of quantity, and higher marginal prices in higher tiers promote equity. These effects work against the assumption of using income as a proxy for consumption in rate design, making IBP an unreliable tool for achieving distributional goals. This finding aligns with previous literature concluding that IBP changes often favor higher-income households (Ruijs, Zimmermann and van den Berg (2008), Echeverri (2023), Wichman (2024)).³³

C. Shadow Cost of the Policy Constraints

The counterfactual analysis proves that, when weather shifts the demand curve right, it is difficult to improve or even maintain distributional equity while simultaneously achieving conservation goals. For the lowest-income stratum, this welfare imbalance stems from both the policy constraint and the extreme weather itself. While the weather is fully exogenous and the utility cannot control its welfare impact, policy constraints, on the other hand, can be mitigated by the utility. To isolate the welfare loss that can be mitigated by the utility, I first calculate the welfare impact of extreme weather alone by applying the status quo price (p_0) to all counterfactual weather scenarios. The difference between that result and the welfare under the optimal price represents the welfare effect of the optimal pricing with their policy constraints. I perform this exercise specifically for the lowest-income stratum. The results for all scenarios can be found in the 'Status

³²See the figure in the 'Price Elasticities of Income Strata' section of the Online Appendix.

³³Both Ruijs, Zimmermann and van den Berg (2008) and Wichman (2024) have pointed out that IBP itself does not effectively target the lower stratum and that more active progressive measures are needed.

Quo Price - Weather Change' section of the Online Appendix.

For the mean-shift scenarios (ζ_1), when $\zeta_1 < 0$, the binding conservation constraint generates a shadow cost that reduces welfare. In the driest condition ($\zeta_1 = -0.25$), welfare under the status quo price is -2.41% on average. Compared to the optimal price result (-4.49%) from Figure 8, this implies the shadow cost of the conservation constraint incurs an average welfare loss of \$74.2 per household per month for the lowest stratum. When $\zeta_1 > 0$, even though the revenue loss function is constructed as a policy constraint, the less strictness of following this constraint in reality made the optimal pricing and its distributional effect work as intended, creating a welfare increase from 2.18% to 3.52% , which is \$45.3 per household per month for the lowest stratum.

For the variance-shift scenarios (ζ_2), the effects are similar. In the lowest variance condition ($\zeta_2 = 0.75$), welfare under the status quo price is 2.45% . Compared to the optimal price result (3.88%) from Figure 9, this implies that a less strict revenue constraint enables the optimal pricing to create a welfare gain of \$48.5 per household per month for the lowest stratum. In the highest variance condition ($\zeta_2 = 1.25$), welfare under the status quo price is -2.48% . Compared to the optimal price result (-4.60%) from Figure 9, this implies a shadow cost from the conservation constraint of \$74.9 per household per month.

Both shifting mean and shifting variance showcase that the effectiveness of the optimal pricing procedure with accounting for welfare equity can be hindered by the strictness of the policy constraints. I modeled the policy behaviors of the utility using data and its goals. With the dwindling water supply, more metropolitan areas in the southwest regions of the US will be like Austin Water, adopting more and more strict conservation policies. Revenue, on the other hand, could be savaged by the city government as typically, the utility company is a public company. If the utility cares more about its revenue feasibility, i.e., λ is higher, a more strict revenue constraint could also impose a welfare burden for the lower-income households. In addition, I also check the robustness of the result with different segments of built years to ensure the estimated income from house appraisal value won't bias the result.

D. Zeroscaping for the Lowest Income Stratum

The preceding analysis concludes that under extreme weather, policy constraints on optimal prices cause greater welfare losses for the lowest-income stratum. Since these constraints are essential for utility operations, mitigating this welfare imbalance requires an additional policy targeted at this group. IBP's failure to achieve the distributional goal is largely due to the mismatch between income level and consumption level. To fully understand what variables could predict the consumption heterogeneity, I performed a reduced-form analysis (see Table Logit Regression Model of Predicting High Quantity for Low-Income Household in the Online Appendix, which shows that factors like house size, number of bathrooms, and vegetation health (NDVI) are significant predictors of high usage. Of these, NDVI becomes most pronounced during periods of low precipitation. Therefore, a natural policy to test is zeroscaping—reducing NDVI.

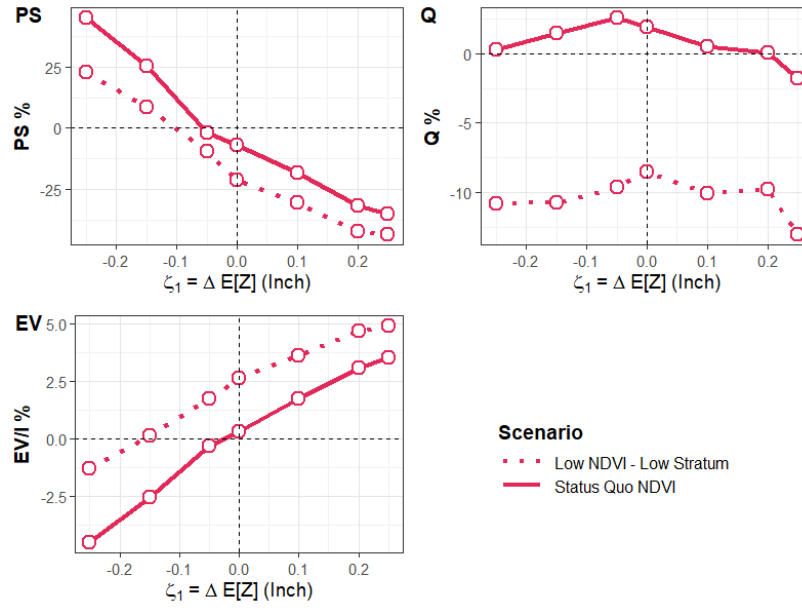
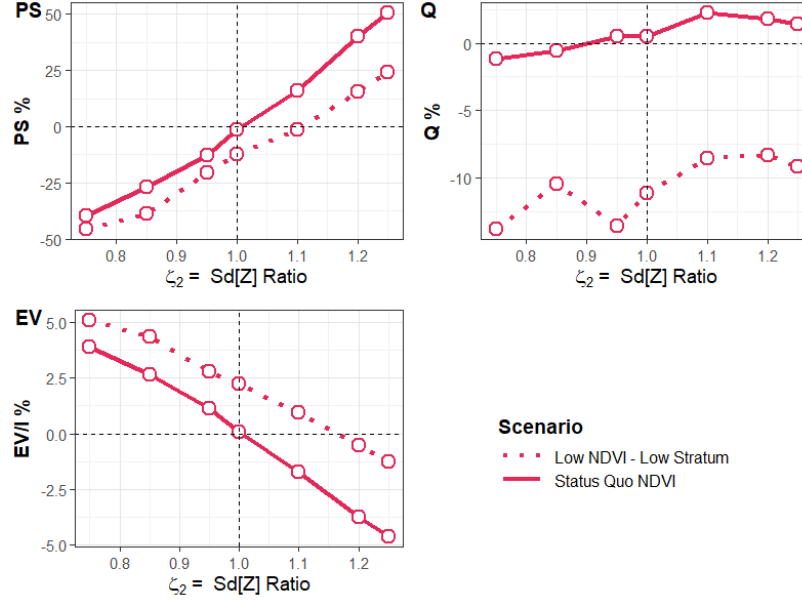
For the counterfactual, I halve the status quo NDVI for the lowest-income stratum, simulating the removal of roughly half their lawns,³⁴ and repeat the optimal pricing exercise. Since NDVI only changes for the lowest stratum, I present only their welfare changes.

Unsurprisingly, by halving NDVI, the lowest stratum consumes and pays less, resulting in an increase in welfare across all weather conditions, particularly when $\zeta_1 < 0$ and the demand curve shifts rightward. In the most extreme dry condition ($\zeta_1 = -0.25$), zeroscaping improves EV/I for the lowest stratum from -4.49% to -1.29% , corresponding to a gain of roughly \$111.82 per household per month. This measures the immediate welfare effect, not including long-term benefits.³⁵

The pattern is similar for variance shifts. When demand shifts rightward ($\zeta_2 > 1$), xeriscaping improves welfare for the lowest stratum. In the extreme case of $\zeta_2 = 1.25$, it improves EV/I from -4.60% to -1.26% , corresponding to roughly \$116.39 per household per month.

³⁴Note that I only modified $NDVI > 0$; if the status quo NDVI was already negative, I did not change it.

³⁵A one-time investment in zeroscaping would provide a near-permanent downward shift in the demand curve, increasing future welfare.

Figure 10. : Welfare Comparison of Halving NDVI ($\zeta_1 \in [-0.25, 0.25]$)Figure 11. : Welfare Comparison of Halving NDVI ($\zeta_2 \in [0.75, 1.25]$)

To further investigate the value of this policy, I implemented multiple NDVI reduction levels, from 0 to 0.9 (representing a 90% lawn reduction). Figure 12 shows the welfare change for the two most extreme rightward-shift scenarios.

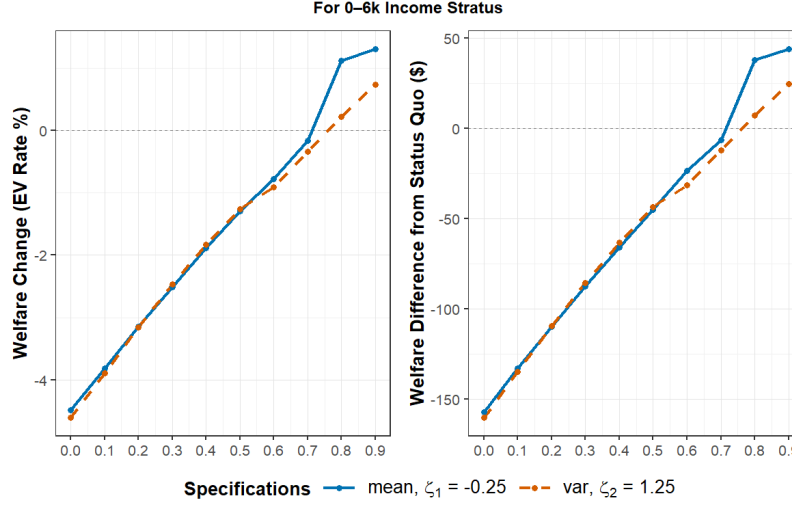


Figure 12. : Welfare Result under Extreme Weather Conditions by NDVI reduction level $\in [0, 0.9]$

The graph shows a clear increasing trend in welfare as the NDVI reduction level increases. For $\zeta_1 = -0.25$, a 0.3 reduction in NDVI generates \$69.53 per household per month, almost nullifying the welfare loss from the conservation constraint's shadow cost. A 0.7 reduction nullifies the welfare losses from both the shadow cost and the extreme weather itself. For $\zeta_2 = 1.25$, welfare is on par with the previous case up until 0.7, where zeroscaping will generate higher welfare for $\zeta_1 = -0.25$ when the reduction level is larger than 0.7. Regardless of this difference, removing water-intensive vegetation by 70% will make sure both weather and policy won't impact lower-income households disproportionately, and zeroscaping remains a viable policy solution to improve the distributional effect.

VI. Conclusion

In this paper, I estimate optimal Inclining Block Price (IBP) tariffs for water utilities facing extreme precipitation volatility. Using a structural Discrete-Continuous Choice model incorporating satellite-derived vegetation indices (NDVI) to proxy outdoor water usage, I show that high-income households exhibit greater consumption volatility and price sensitivity during dry periods. This mismatch between household income and consumption severely undermines the distributional objectives of standard IBP structures.

Through an empirical Ramsey pricing model with revenue and conservation constraints, I evaluate optimal tariffs across shifting weather patterns. When drought or high volatility pushes demand rightward, binding conservation constraints compound the welfare losses induced by the weather itself. By decomposing these effects, I find the shadow cost of these policy constraints falls almost exclusively on the lowest-income stratum, imposing an average welfare loss of \$74.2 and \$74.9 per month during extreme droughts and high-variance periods, respectively. This confirms that price alone is an insufficient instrument to simultaneously achieve conservation and equity goals.

Since pricing mechanisms fail to protect vulnerable consumers during demand surges, complementary policies are essential. Simulations show that zeroscaping—reducing low-income lawn vegetation (NDVI) by 30%—generates \$72.07 in monthly welfare for these households, nullifying the conservation constraint’s regressive shadow cost. Future work will endogenize landscaping decisions and model how low-income households use smoothing programs like ‘Budget Billing’ to insure against weather stochasticity.

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